

CHERLAK, Russian Federation

WMO: 287990

Lat: 54.167N

Lon: 74.800E

Elev: 376

StdP: 14.50

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-23.3	-32.9	0.9	-28.0	-28.4	1.1	-23.3	16.8	14.6	14.8	16.0	3.1	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	87.8	66.6	84.6	65.9	81.4	64.5	69.4	82.0	67.9	79.7	66.4	77.9	6.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	95.2	73.9	63.6	89.1	72.7	62.1	84.4	71.8	33.7	82.5	32.5	79.8	31.3	78.0	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.9	13.7	11.7	DB	-32.1	93.2	6.3	2.7	-36.7	95.1	-40.4	96.7	-43.9	98.3	-48.5	100.2	(4)
			WB	-32.5	72.5	6.2	3.0	-36.9	74.7	-40.6	76.4	-44.0	78.1	-48.5	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	-0.7	3.6	18.8	40.4	55.7	65.2	67.8	63.5	52.5	39.4	20.2	6.0
	DBStd	26.58	13.67	12.46	11.12	10.98	9.41	7.82	6.64	7.15	8.18	8.73	13.38	13.84
	HDD50	6880	1572	1298	967	323	50	3	0	1	64	344	893	1365
	HDD65	10848	2037	1718	1432	737	314	92	50	117	383	795	1343	1830
	CDD50	1848	0	0	0	36	226	460	551	420	140	15	0	0
	CDD65	338	0	0	0	0	25	99	135	70	9	0	0	0
	CDH74	3325	0	0	0	11	347	980	1213	656	118	0	0	0
	CDH80	981	0	0	0	1	85	306	375	192	22	0	0	0
Wind	WSAvg	5.3	5.3	5.5	5.9	6.1	5.6	4.6	4.2	4.2	4.6	5.4	6.0	5.8
Precipitation	PrecAvg	14.4	0.8	0.7	0.7	0.7	0.9	1.7	2.5	1.9	1.0	1.2	1.2	1.0
	PrecMax	21.1	1.5	1.1	1.9	2.1	3.2	3.5	5.7	5.1	2.8	2.9	2.9	2.1
	PrecMin	8.9	0.1	0.1	0.2	0.1	0.1	0.2	0.4	0.2	0.1	0.2	0.3	0.3
	PrecStd	3.0	0.4	0.3	0.5	0.5	0.6	0.8	1.6	1.1	0.6	0.7	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.6	44.9	74.6	86.8	90.6	92.7	90.6	82.8	68.2	46.2	34.1
		MCWB	32.0	32.4	38.8	56.6	63.0	65.5	68.0	66.2	62.5	53.2	43.1	32.8
	2%	DB	28.3	30.2	39.1	68.6	81.8	86.9	87.8	85.1	77.2	62.6	42.5	32.0
		MCWB	27.5	29.1	35.6	53.2	60.5	65.8	67.5	66.8	59.4	51.1	39.8	31.2
	5%	DB	24.6	26.7	35.9	63.8	77.8	83.5	84.2	80.8	71.5	57.5	39.3	29.0
		MCWB	23.8	25.7	33.5	50.8	58.5	65.2	66.9	64.6	57.2	48.1	36.9	28.0
	10%	DB	20.5	22.6	33.9	58.4	73.1	79.7	80.7	77.1	66.8	53.1	35.8	24.8
		MCWB	19.7	21.6	32.0	48.1	56.4	63.7	66.1	63.4	55.0	46.0	33.7	23.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	32.6	39.6	58.3	64.9	70.2	71.7	70.1	65.8	55.0	44.2	33.1
		MCDB	33.4	33.3	44.1	72.3	82.2	85.4	82.4	82.4	81.3	65.0	46.0	33.6
	2%	WB	27.6	29.1	36.0	54.3	62.3	68.1	70.0	68.1	61.0	52.3	40.0	31.2
		MCDB	28.3	30.1	38.4	65.5	78.9	81.1	81.9	80.4	72.3	60.6	42.0	32.0
	5%	WB	23.7	25.7	34.0	51.3	59.8	66.4	68.5	66.5	58.3	49.3	37.0	27.9
		MCDB	24.5	26.6	35.7	62.4	74.5	79.1	80.1	77.5	69.8	56.7	39.1	28.9
	10%	WB	19.6	21.5	32.0	48.5	57.6	64.6	67.1	64.9	55.6	46.3	33.8	23.7
		MCDB	20.3	22.5	33.8	58.0	70.4	77.4	78.3	74.9	65.3	52.6	35.5	24.6
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.7	14.3	17.4	21.2	19.4	18.0	18.5	18.3	14.2	11.1	13.0
		MCDBR	15.2	12.8	12.5	26.0	27.3	24.7	22.9	24.3	25.6	23.0	11.6	13.4
		MCWBR	15.2	12.7	10.3	14.6	12.0	10.0	8.5	10.5	12.4	13.4	9.8	13.5
	5% WB	MCDBR	15.0	12.7	12.2	24.2	24.6	21.1	19.5	20.6	23.4	20.8	11.0	13.7
		MCWBR	15.0	12.5	10.7	14.5	12.3	9.7	8.6	10.2	12.3	12.8	9.8	13.6
Clear-Sky Solar Irradiance	taub	0.289	0.319	0.378	0.405	0.404	0.412	0.435	0.428	0.369	0.352	0.328	0.292	
	taud	2.045	2.013	1.932	2.148	2.219	2.280	2.211	2.196	2.320	2.250	2.106	2.009	
	Ebn at Noon	198	234	244	259	267	266	256	249	252	224	180	165	
	Edh at Noon	23	33	45	42	42	40	42	40	31	26	22	19	
All-Sky Solar Radiation	RadAvg	338	680	1169	1477	1823	1944	1774	1463	1065	553	263	218	
	RadStd	24	27	92	118	162	121	124	111	92	74	34	27	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page